

MNYMO Competition 2024 Individual Solutions

MN YOUTH MATH OUTREACH

August 2024

Problem 1. Jinu Lee is going on a diet. He goes to the local fruit market and notices that three apples and four oranges together cost \$12, while four apples and two oranges together cost \$11. How many cents does one apple cost?

Proposed by: Austin Wang

Answer: 200

Solution: Let the cost of an apple be a dollars and let the cost of an orange be o dollars. Then we get the following system of equations

$$\begin{aligned}3a + 4o &= 12 \\4a + 2o &= 11\end{aligned}$$

from which we obtain $a = 2$. Thus, an apple costs 2 dollars, or 200 cents.

Problem 2. Find the area of a rectangle with one side of length 4 and diagonals of length 5.

Proposed by: Austin Wang

Answer: 12

Solution: The area can be calculated as the product of adjacent sides. Since the diagonals have length 5, the other side length is $\sqrt{5^2 - 4^2} = 3$ by the Pythagorean Theorem. Thus, the area is $4 \cdot 3 = \text{12}$.

Problem 3. Ben is finding the square root of a three digit perfect square x . In doing so, however, he accidentally swaps the tens and units digits of x and gets an answer that is one more than the correct answer. Find x .

Proposed by: Austin Wang

Answer: 169

Solution:

Notice that $13^2 = 169$ and $14^2 = 196$. One can verify that these are the only numbers that satisfy the given conditions. Thus, $x = \text{169}$.

Problem 4. How many of the first 200 positive integers are multiples of 5 but not 6?

Proposed by: Austin Wang

Answer: 34

Solution: By the Principle of Inclusion-Exclusion, this can be calculated as the number of multiples of 5 minus the number of multiples of 30, since 30 is the least common multiple of 5 and 6. Thus, we get $\left\lfloor \frac{200}{5} \right\rfloor - \left\lfloor \frac{200}{30} \right\rfloor = 40 - 6 = \text{34}$.

Problem 5. Sam, Gary, and Owen decide to share fruits. Sam has 50 strawberries, Gary has 40 grapes, and Owen has 30 oranges to begin with. Given that any type of fruit can only be given in groups of 6, and every person has all 3 types of fruit at the end, what is the maximum number of fruits Gary could have at the end of the exchange? (Note: Not all the fruit someone initially has must be given out.)

Proposed by: Samuel Kretzschmar

Answer: 88

Solution: First, note that there are $50 + 40 + 30 = 120$ fruits in total. To maximize Gary's fruits, we try to minimize Sam's and Owen's. For Sam to have all three fruits, he must have a group of 6 grapes and a group of 6 oranges, as well as at least $50 - 6 \cdot 8 = 2$ strawberries left from giving all he can, for a total of 14 fruits. Similarly, Owen must have a group of 6 strawberries and a group of 6 grapes. If Owen gives away as many oranges as he can, he would be left with $30 - 6 \cdot 5 = 0$; to make sure he has some left over, he has a minimum of 6 oranges, leading to a total of $6 + 6 + 6 = 18$ fruits. Thus Gary has a maximum of $120 - 14 - 18 = 88$ fruits, with $30 - 6 - 6 = 18$ oranges, $40 - 6 - 6 = 28$ grapes, and $50 - 2 - 6 = 42$ strawberries.

Problem 6. Find the sum of the prime factors of the number $S = 4^8 - 1$.

Proposed by: Ryan Niu

Answer: 282

Solution: Using the difference of squares repeatedly, we have

$$4^8 - 1 = (4^4 + 1)(4^4 - 1) = (4^4 + 1)(4^2 + 1)(4^2 - 1) = (4^4 + 1)(4^2 + 1)(4 + 1)(4 - 1) = (257)(17)(5)(3).$$

This means the sum of the prime factors is simply $257 + 17 + 5 + 3 =$ 282.

Problem 7. How many ways are there to color the vertices of a tetrahedron (a 3D figure made up of 4 vertices, 4 triangular faces, and 6 edges), where each vertex can be colored with one of six colors, such that vertices connected by an edge have distinct colors? Colorings that differ by rotations and/or reflections are considered distinct.

Proposed by: Angie Huang

Answer: 360

Solution: Each of the vertices are pairwise connected by an edge, so the vertices must all have distinct colors. Thus, our answer is $6 \cdot 5 \cdot 4 \cdot 3 = 360$.

Problem 8. Alexander Ren and Sam Kretzschmar are both listening to the same song on repeat. However, Sam accidentally set it to play at 1.25 times speed, so for every one second he listens to the song, 1.25 seconds of the original song has elapsed. Given that the song is 60 seconds long and they both start playing it at the same time, how many seconds does it take for Sam and Alex to start playing the song at the same time again?

Proposed by: Sneha Kundu

Answer: 240

Solution: It takes 60 seconds for Alex to listen to it once and $\frac{1}{1.25} \cdot 60 = 48$ seconds for Sam to listen to it once, so they start at the same time after $\text{lcm}(60, 48) = 240$ seconds.

Problem 9. Integers x and y satisfy the equation $xy = x - y + 5$. Find the greatest possible value of $x + y$.

Proposed by: Austin Wang

Answer: 5

Solution: Notice that the equation can be rewritten as $xy - x + y = 5$. Using Simon's favorite factoring trick, we can subtract 1 from both sides and factor to get $(x + 1)(y - 1) = 4$. We want to maximize $x + y$. Notice that $x + y = (x + 1) + (y - 1)$. This is maximized when $x + 1 = 1$ and $y - 1 = 4$ or $x + 1 = 4$ and $y - 1 = 1$. Either way, we get a sum of 5.

Problem 10. Four spheres of radius 1 are arranged such that their centers form the vertices of a square of side length 2. A larger sphere is placed around these four spheres such that the larger sphere is internally tangent to each smaller sphere. Find the smallest possible value of the radius of the larger sphere.

Proposed by: Ryan Niu

Answer: $\sqrt{2} + 1$

Solution: Consider viewing the figure from a bird's eye view. Now, our diagram is simply 4 circles of radius 1 inscribed in a larger circle. By symmetry, we can see that the center of the square formed

by the circles is also the center of the outer circle, so from one corner of the square to the center is $\sqrt{2}$ by Pythagorean's Theorem, and an extension of 1 will give us the radius of the larger circle. Thus, the radius of our larger sphere is $\boxed{\sqrt{2} + 1}$.

Problem 11. An integer k is considered to be *level* n if k ends in exactly n 0's. For example, the number 100 is level 2. What is the smallest integer k such that $k!$ is level 10?

Proposed by: Austin Wang

Answer: $\boxed{45}$

Solution: The factorial $n!$ of a number n is computed as $n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1$. Consider that the number 10 is computed through the product of 2 and 5 or simply a factor of 10 itself. This means that a factorial of every multiple of 5 has an extra digit of 0 (holds linearly until 50). As the number 5 is level 1, we count up the multiples of 5 to find that $\boxed{45}$ is level 10.

Problem 12. Find the sum of all real numbers m such that the polynomial $P(x) = x^2 + mx + m + 2024$ has exactly one distinct root.

Proposed by: Austin Wang

Answer: $\boxed{4}$

Solution: P has one distinct root if and only if its discriminant is equal to 0. Therefore, we have

$$m^2 - 4m - 8096 = 0.$$

. Notice that we only have to find the *sum* of all such m . We can use Vieta's on the above quadratic to find that the sum of all m is $-\left(\frac{-4}{1}\right) = \boxed{4}$.

Remark. One can find that the values of m are -88 and 92 .

Problem 13. Call a number *vanilla* if at least half of its digits are the same, and call a number *unique* if all of its digits are distinct from each other. What is the ratio of the number of vanilla 4-digit numbers to the number of unique 4-digit numbers?

Proposed by: Austin Wang

Answer: $\boxed{\frac{62}{63}}$

Solution: Notice that a vanilla four digit number has at least 2 digits that are the same. This means that every four digit number is either vanilla or unique. There are $9 \cdot 10 \cdot 10 \cdot 10 = 9000$ four digit numbers, $9 \cdot 9 \cdot 8 \cdot 7 = 4536$ of which are unique. Thus, there are $9000 - 4536 = 4464$ vanilla numbers, so the ratio is

$$\frac{4464}{4536} = \boxed{\frac{62}{63}}.$$

Problem 14. Ricky flips a fair coin five times, and Miguel flips a fair coin three times. What is the probability that Ricky lands more heads than Miguel?

Proposed by: Austin Wang

Answer: $\boxed{\frac{163}{256}}$

Solution: We can do casework on the number of heads that Ricky lands.

Case 1: 1 head

If Ricky lands 1 head, Miguel must land 0 heads. There are $\binom{5}{1} \cdot 1 = 5$ ways in this case.

Case 2: 2 heads

If Ricky lands 2 heads, Miguel must land either 0 or 1 heads. There are $\binom{5}{2} \cdot (1 + \binom{3}{1}) = 40$ ways in this case.

Case 3: 3 heads

If Ricky lands 3 heads, Miguel must land either 0, 1, or 2 heads. There are $\binom{5}{3} \cdot (1 + \binom{3}{1} + \binom{3}{2}) = 70$ ways in this case.

Case 4: 4 or more heads

If Ricky lands 4 or 5 heads, he is guaranteed to land more than Miguel. The number of ways here is $\binom{5}{4} + \binom{5}{5} \cdot 2^3 = 48$.

Thus, the total number of valid sequences is 163. There are $2^5 \cdot 2^3 = 256$ possible sequences, for an overall probability of $\frac{163}{256}$.

Problem 15. Consider the polynomial $p(x) = x^{100} + 2x^{99} + \cdots + 100x + 101$. Let the roots of $p(x)$ be $r_1, r_2, \dots, r_{100}$. Compute $\sum_{i=1}^{100} \frac{1}{r_i}$.

Proposed by: Austin Wang

Answer: $\boxed{-\frac{100}{101}}$

Solution: To begin, observe that $\sum_{i=1}^{100} \frac{1}{r_i}$ can be written as a single fraction with the numerator as the sum of the products of the roots taken 99 at a time. The denominator of this fraction is the product of all 100 roots. Through observation of the expansion of $p(x) = (x - r_1)(x - r_2) \cdots (x - r_{100})$ or Vieta's formulas, we get that the product of the roots is 101 and the sum of the products of the roots taken 99 at a time is -100 . Thus, the answer becomes $\boxed{-\frac{100}{101}}$.

Problem 16. Let ABC be a triangle, and let D and E be points on sides BC and AC , respectively, such that $BD : DC = 1 : 2$ and $AE : EC = 1 : 3$. Let lines $\overline{AD}$ and $\overline{BE}$ intersect at P , and let line $\overline{CP}$ intersect side AB at point F . Compute CP/PF .

Proposed by: Austin Wang

Answer: $\boxed{5}$

Solution: Set up a mass points system. Let the mass of point A be 3. Using the given ratios, this implies that the mass of point C is 1, which implies that the mass of point B is 2. Thus, the mass of point F is $3 + 2 = 5$. Since the mass of point C is 1, we can conclude that the ratio CP/PF is $\frac{5}{1} = \boxed{5}$.

Problem 17. In each of its moves, the *downright frog* moves from its current point (x, y) to $(x + 1, y - 2)$ or to $(x + 2, y - 1)$. Compute the number of ways for the downright frog to move from $(0, 15)$ to $(15, 0)$.

Proposed by: Ryan Niu

Answer: $\boxed{252}$

Solution: In total, the frog needs to travel 15 spaces downwards and 15 spaces rightwards, so to satisfy this we must have an equal amount of the two possible moves. Each move covers 3 of the 30 spaces in total length, so we will have ten moves total, with 5 of each move. Thus, the total number of ways to order these moves is simply $\binom{10}{5} = \boxed{252}$.

Problem 18. Points A, B, C, D , and E lie on a circle with radius 5, in that order. Given that $\angle ABC = 90^\circ$, $CD = 6$, and $\cos(\angle ABE) = \frac{2}{3}$, the length of DE can be written in the form $\frac{a-b\sqrt{c}}{d}$, where a, b, c , and d are positive integers such that c is squarefree and $\gcd(a, b, d) = 1$. What is $a + b + c + d$?

Proposed by: Angie Huang

Answer: $\boxed{30}$

Solution: Given that $\angle ABC = 90^\circ$, we know that AC is the diameter of the circle. Thus, $\triangle ADC$ is right with $AD = 8$. Additionally, $\triangle AEC$ is right with $AE = 10 \sin(\angle ACE) = 10 \sin(\angle ABE) = 10\sqrt{1 - \cos^2(\angle ABE)} = \frac{10\sqrt{5}}{3}$. Finally, since $\angle ADE = \angle ABE$, Law of Cosines on $\triangle ADE$ gives

$$\begin{aligned} AE^2 &= AD^2 + DE^2 - 2AD \cdot DE \cos(\angle ADE) \\ \frac{500}{9} &= 64 + DE^2 - 16DE \cdot \frac{2}{3} \\ DE^2 - \frac{32}{3}DE + \frac{76}{9} &= 0 \\ DE &= \frac{\frac{32}{3} \pm \sqrt{\frac{1024}{9} - 4 \cdot \frac{76}{9}}}{2} = \frac{\frac{32}{3} \pm 4\sqrt{5}}{2} = \frac{16 - 6\sqrt{5}}{3}. \end{aligned}$$

Therefore, $16 + 6 + 5 + 3$ gives us $\boxed{30}$. Note that we take the smaller, positive solution as E must lie on the same side of AC as D .

Problem 19. The real number x satisfies the equation $x + \frac{1}{x} = 3$. The value of $x^{2024} + \frac{1}{x^{2024}}$ can be written as $\frac{m}{2^{2024}}$. Find the remainder when m is divided by 9.

Proposed by: Ryan Niu

Answer: $\boxed{8}$

Solution: Directly solve for x , and we obtain $x = \frac{3 \pm \sqrt{5}}{2}$. Then if we denote $S_n = x^n + \frac{1}{x^n}$ we find that conveniently enough, $S_n = (\frac{3+\sqrt{5}}{2})^n + (\frac{3-\sqrt{5}}{2})^n$. Now, it is clear that m is simply $(3+\sqrt{5})^{2024} + (3-\sqrt{5})^{2024}$. Using binomial expansions, we combine to get

$$\sum_{k=0}^{2024} \binom{2024}{k} \cdot 3^{(2024-k)} \cdot (\sqrt{5})^k + \sum_{k=0}^{2024} \binom{2024}{k} \cdot 3^{(2024-k)} \cdot (-\sqrt{5})^k = 2 \left(\sum_{j=0}^{1012} \binom{2024}{2j} \right) \cdot 3^{(2024-2j)} \cdot (\sqrt{5})^{2j}.$$

In the end, these terms will all have a factor of 9 except for the final term $2 \cdot (\sqrt{5})^{2024} = 2 \cdot 5^{1012}$, so we simply have to consider this number modulo 9. We know $5^2 \equiv -2 \pmod{9}$ and $2^3 \equiv -1 \pmod{9}$ so, $2 \cdot 5^{1012} \pmod{9} \equiv 2 \cdot (-2)^{506} \pmod{9} \equiv 2^{507} \pmod{9} \equiv (-1)^{169} \pmod{9} \equiv \boxed{8} \pmod{9}$.

Problem 20. A group of 16 people stand equally spaced in a circle, facing inwards. Alice and Bob stand opposite one another in the circle, and Bob is handed a ball. Every second, the person holding the ball gives it to the person to their left with probability $\frac{2}{5}$ and the person to their right with probability $\frac{3}{5}$. If the probability that Alice is first given the ball from the person to her right is $\frac{m}{n}$, for relatively prime positive integers m and n , find $m + n$.

Proposed by: Samuel Kretzschmar

Answer: $\boxed{7073}$

Solution:

Label the people to the left of Alice in order with $A_1, A_2, \dots, A_{15}$, and the probability of Alice getting the ball from A_{15} (the person to her right) given the ball is at A_n as $P(A_n)$. Also, for convenience, say that A_0 represents Alice from the right of A_1 , and A_{16} represents Alice from the left of A_{15} ; thus $P(A_0) = 0$ and $P(A_{16}) = 1$. Note that Bob is at A_8 .

Now, the probabilities yield the formula

$$P(A_n) = \frac{3}{5}P(A_{n-1}) + \frac{2}{5}P(A_{n+1})$$

for integers $1 \leq n \leq 15$. Now, assume that $P(A_1) = x$. Checking some values using the recursion returns $P(A_2) = \frac{5}{2}x$, $P(A_3) = \frac{19}{4}x$, etc. We note that the values seem to follow the pattern $P(A_n) = \left(1 + \frac{3}{2} + \left(\frac{3}{2}\right)^2 + \dots + \left(\frac{3}{2}\right)^{n-1}\right)x = 2 \left(\left(\frac{3}{2}\right)^n - 1\right)x$, which can be easily checked as true with induction as

$$\begin{aligned} \frac{3}{5} \cdot 2 \left(\left(\frac{3}{2}\right)^{n-1} - 1\right)x + \frac{2}{5} \cdot 2 \left(\left(\frac{3}{2}\right)^{n+1} - 1\right)x &= \\ 2 \left(\frac{3}{5} \cdot \frac{2}{3} + \frac{2}{5} \cdot \frac{3}{2}\right) \left(\frac{3}{2}\right)^n x - 2x &= 2 \left(\left(\frac{3}{2}\right)^n - 1\right)x. \end{aligned}$$

Now, using this, we have $P(A_{16}) = 2 \left(\left(\frac{3}{2}\right)^{16} - 1\right)x = 1$ so $x = \frac{1}{2 \left(\left(\frac{3}{2}\right)^{16} - 1\right)}$. Thus

$$P(A_8) = \frac{2 \left(\left(\frac{3}{2}\right)^8 - 1\right)}{2 \left(\left(\frac{3}{2}\right)^{16} - 1\right)} = \frac{2^8(3^8 - 2^8)}{3^{16} - 2^{16}} = \frac{2^8}{2^8 + 3^8}.$$

As $\gcd(2^8 + 3^8, 2^8) = \gcd(3^8, 2^8) = 1$, the answer is $2 \cdot 2^8 + 3^8 = 512 + 6561 = 7073$.

Problem 21 (Tiebreaker). Define a *limited number* to be a positive integer which has exactly two distinct digits in its decimal representation (leading zeros are not considered). For example, 3344433 and 20 are limited numbers, but 0111 and 12332 are not. Find the 225th limited number which is divisible by 90. If the answer is A and your solution is N , the lowest value of $\left\lceil \log \left(\frac{N}{A} \right) \right\rceil$ wins.

Proposed by: Samuel Kretzschmar

Answer: 303003000

Solution: Call a limited number divisible by 90 a restricted number. Let a_n be the number of restricted numbers with n digits. Note that any restricted number must end in a 0, so the digits in any restricted number will be 0 and some other integer $1 \leq k \leq 9$. Thus, we can split into three cases, assuming the number of digits is n :

First, consider if $k = 9$. Then any arrangement of 0s and 9s would be divisible by 9 and 10, as long as it ends in a 0 and starts with a 9, which is possible for $n \geq 2$. As every other digit can be chosen as 0 or 9, the number of ways this can be done is 2^{n-2} .

Now, if $k = 3$ or $k = 6$, again the number must begin with k and end with 0, but in order to be divisible by 9, the number of digits that are k must be divisible by 3 to satisfy the divisibility rule for 9. Thus the number in this case is

$$2 \left(\binom{n-2}{2} + \binom{n-2}{5} + \binom{n-2}{8} + \dots \right)$$

as we choose out of the $(n-2)$ middle digits x of them to be k , where x must be 1 less than a multiple of 3, then multiply by the number of possible values of k . Note that this value is only positive if $n \geq 4$.

If k is not divisible by 3, we would have $n \geq 10$ as the number of k -digits would need to be divisible by 9. Thus we can calculate a_n for $n < 9$ without considering this case. We notice that

$$\begin{aligned} a_2 + a_3 + \dots + a_8 &= \sum_{n=2}^8 \left(2^{n-2} + 2 \binom{n-2}{2} + 2 \binom{n-2}{5} + 2 \binom{n-2}{8} + \dots \right) \\ &= \sum_{k=0}^6 \left(2^k + 2 \binom{k}{2} + 2 \binom{k}{5} \right) \\ &= 2^7 - 1 + 2 \binom{7}{3} + 2 \binom{7}{6} \\ &= 127 + 70 + 14 \\ &= 211 \end{aligned}$$

with $\sum_{k=0}^6 2^k = 2^7 - 1$ from finite summation of geometric series and $\sum_{k=0}^6 \left(2 \binom{k}{2} + 2 \binom{k}{5} \right) = 2 \binom{7}{3} + 2 \binom{7}{6}$ using the hockey stick identity. Thus the answer must be the 14th ($225 - 211 = 14$) restricted number with 9 digits, which we can calculate directly by listing permutations* of 3 and 0 which are restricted numbers in order (remembering that there must be $3m$ 3-digits for some positive integer m):

- 1 : 300000330
- 2 : 300003030
- 3 : 300003300
- 4 : 300030030
- 5 : 300030300
- 6 : 300033000
- 7 : 300300030
- 8 : 300300300
- 9 : 300303000
- 10 : 300330000
- 11 : 300333330
- 12 : 303000030

13 : 303000300

14 : 303003000

*Note that this listing can be more easily done by noticing that the first 10 shown permutations are simply permuting the positions of two 3s and three 0s among the 4th, 5th, 6th, 7th, and 8th digits.